University of Central Punjab
Faculty of Information Technology and Computer Science

Week No. 5

Lab Manual: Information Security

Lab Week 5: Web Server Scanning and Testing Vulnerabilities

1. Lab Objectives

- Understand the basic concepts of network scanning.
- Configure two local web servers on Windows.
- Perform port scanning using Zenmap.
- Identify running services and their versions.
- Examine HTTP security configurations.
- Analyze scanning results and identify potential security weaknesses.

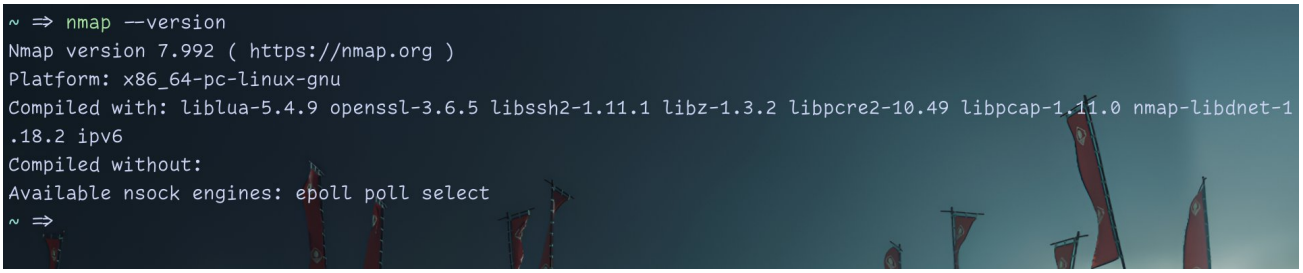
2. Introduction

Network scanning is an important technique in information security. It helps security professionals identify active hosts, open ports, running services, and possible security weaknesses.

Nmap (Network Mapper) is an open-source network scanning tool used for network discovery and security auditing. It allows users to perform network scans without manually executing commands in the command line.

In this lab, students will configure two local web servers and use Zenmap to perform network scanning and basic security assessments.

PART A: WEB SERVER CONFIGURATION



Step 1: Verify Python Installation

1. Open Windows Command Prompt (CMD).
2. Enter the following command:

```
py --version
```

3. Verify that the installed Python version is displayed.

Python will be used to create two local HTTP web servers.

Step 2: Configure Web Server 1

Open Command Prompt and execute the following commands:

Create a directory:

```
mkdir "%USERPROFILE%\WebServer1"
```

Navigate to the directory:

```
cd /d "%USERPROFILE%\WebServer1"
```

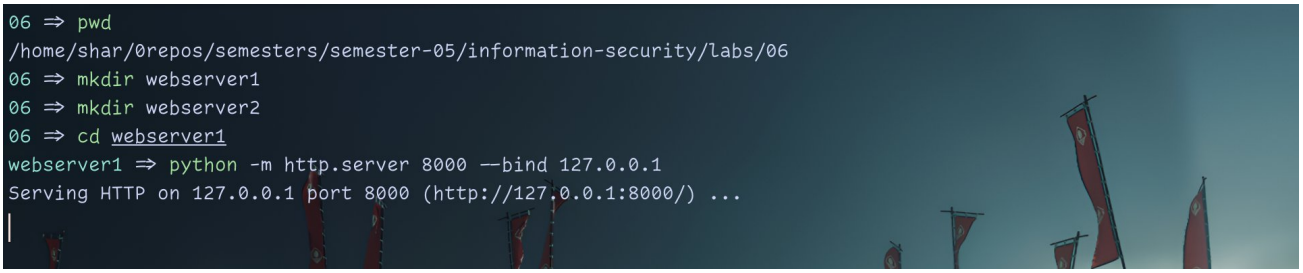
Start the first web server:

```
py -m http.server 8000 --bind 127.0.0.1
```

Expected Output:

```
Serving HTTP on 127.0.0.1 port 8000
```

The first web server is now running on port 8000. Keep this Command Prompt window open.



Step 3: Configure Web Server 2

Open another Command Prompt window and execute the following commands:

```
mkdir "%USERPROFILE%\WebServer2"

cd /d "%USERPROFILE%\WebServer2"
```

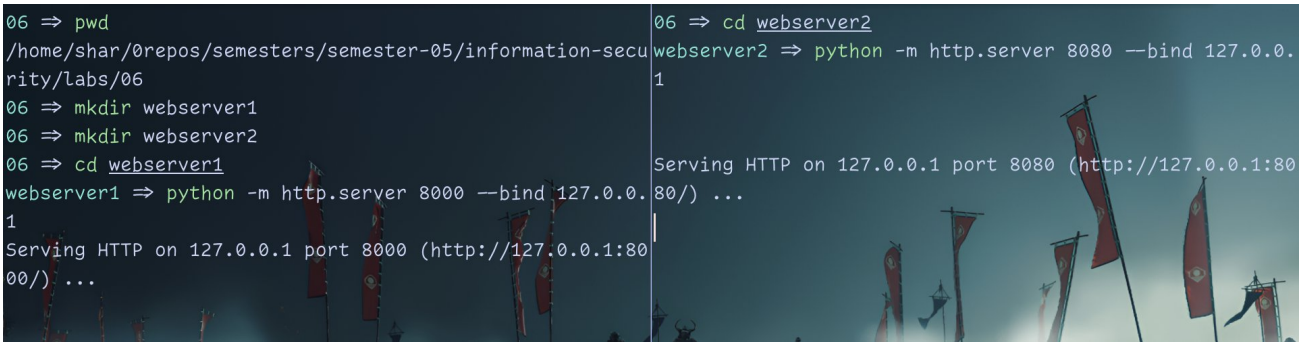
Start the second web server:

```
py -m http.server 8080 --bind 127.0.0.1
```

Expected Output:

```
Serving HTTP on 127.0.0.1 port 8080
```

The second web server is now running on port 8080. Keep both Command Prompt windows open.

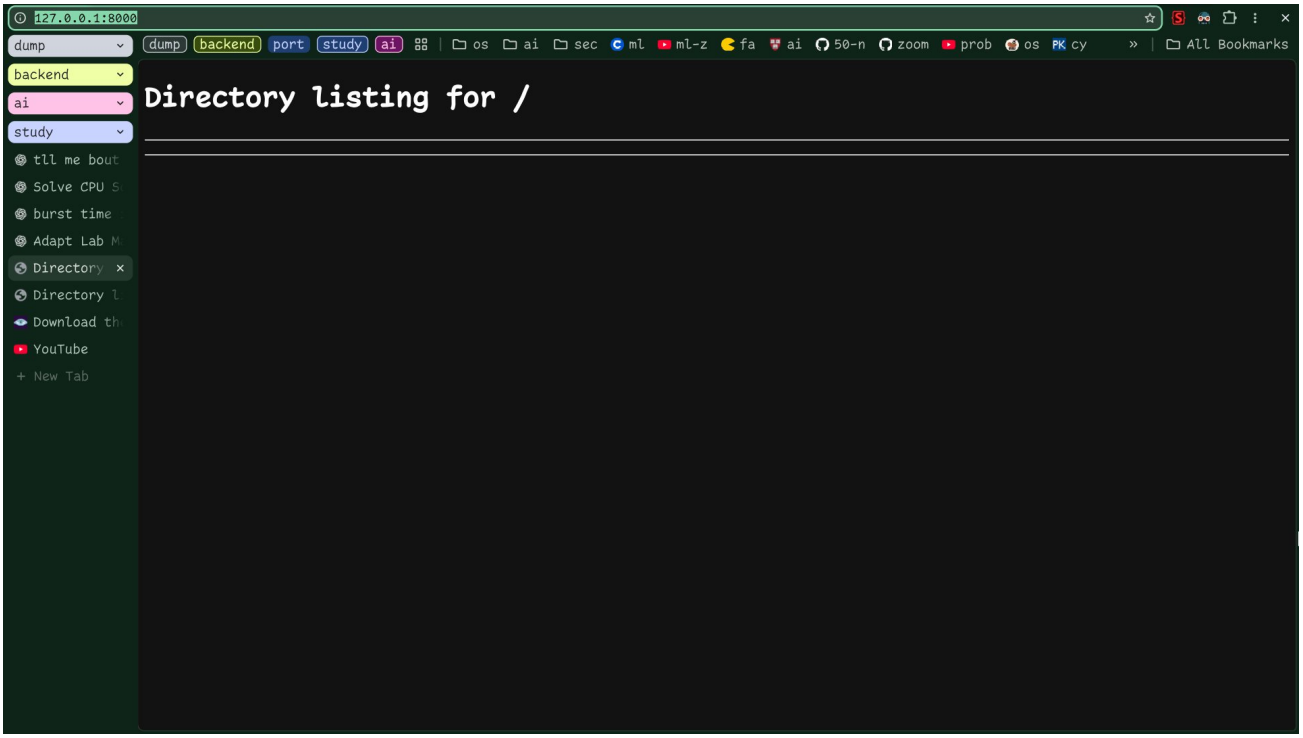


Step 4: Verify Both Web Servers

Open a web browser and enter the following addresses separately:

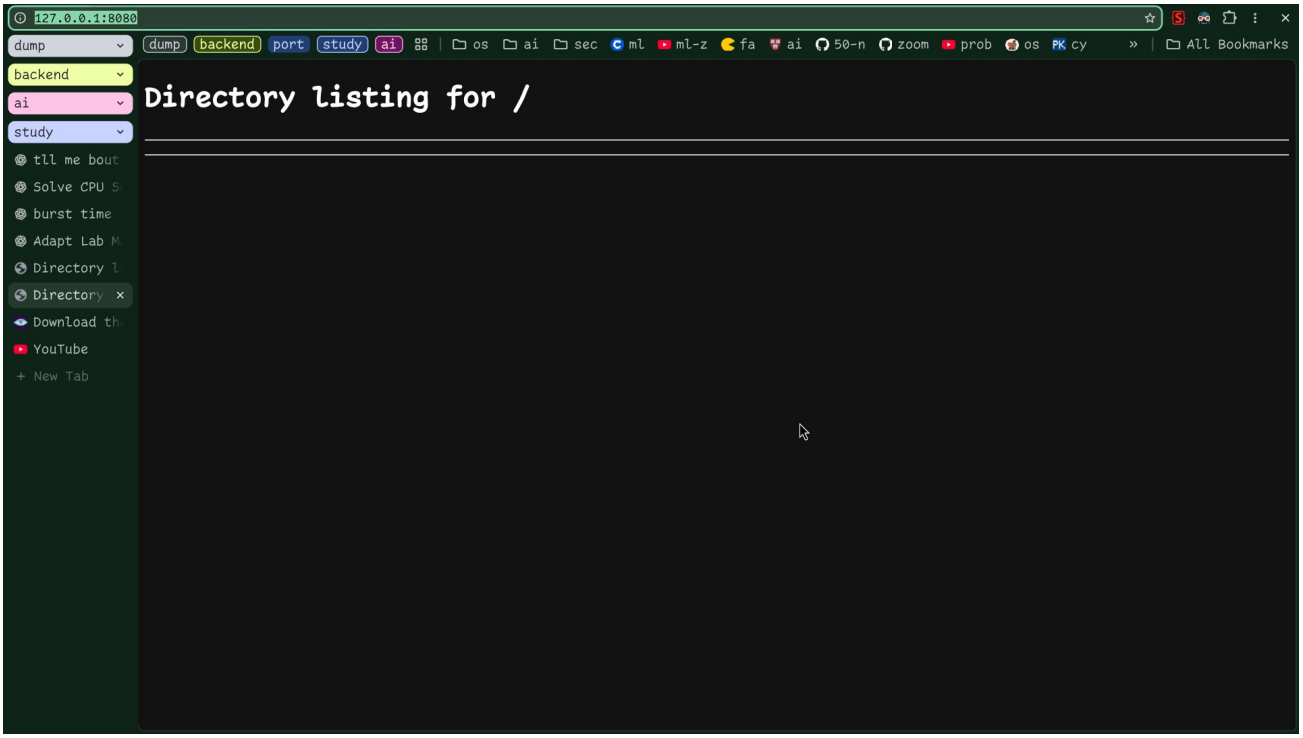
Web Server 1:

```
http://127.0.0.1:8000
```



Web Server 2:

```
http://127.0.0.1:8080
```



Both addresses should display a directory listing or an HTTP response from the Python web server.

PART B: NETWORK SCANNING USING ZENMAP

Step 5: Launch Zenmap

1. Open Zenmap from the Windows Start menu.
2. Locate the Target, Profile, and Command fields.
3. Enter the target IP address:

```
127.0.0.1
```

4. Enter the required scanning command in the Command field.
5. Click the Scan button.

Step 6: Perform Basic Port Scanning

Enter the following command in Zenmap:

```
nmap -sT -p 8000,8080 127.0.0.1
```

Click Scan.

Command Explanation:

- nmap: Starts the network scanning tool.
- -sT: Performs a TCP Connect scan.
- -p: Specifies the ports to scan.
- 8000,8080: Identifies the two target ports.
- 127.0.0.1: Specifies the local computer.

Expected Result:

```
PORT     STATE    SERVICE \n8000/tcp   open     http-alt\n8080/tcp   open     http-proxy
```



The service names may vary depending on the Nmap version and service database.

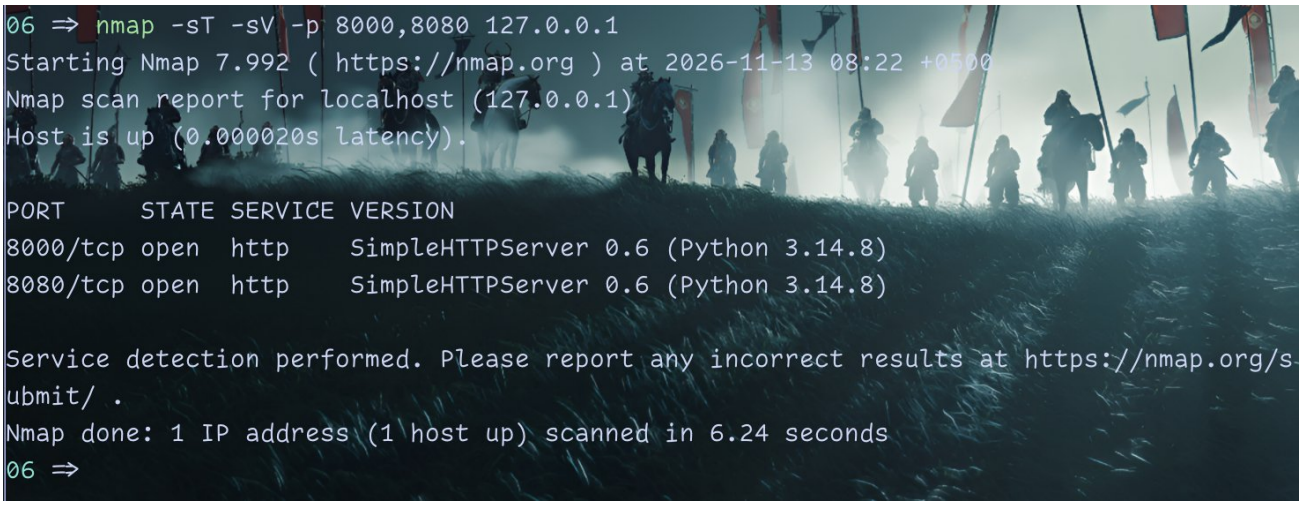
An open port indicates that an application is accepting network connections on that port. A closed port indicates that no application is listening on that port.

Step 7: Perform Service and Version Detection

Enter the following command in Zenmap:

```
nmap -sT -sV -p 8000,8080 127.0.0.1
```

Click Scan.



Command Explanation:

-sV: Attempts to identify running services and their versions.

-p 8000,8080: Scans both web server ports.

PART C: BASIC WEB SERVER SECURITY ASSESSMENT

Step 8: HTTP Methods Scanning

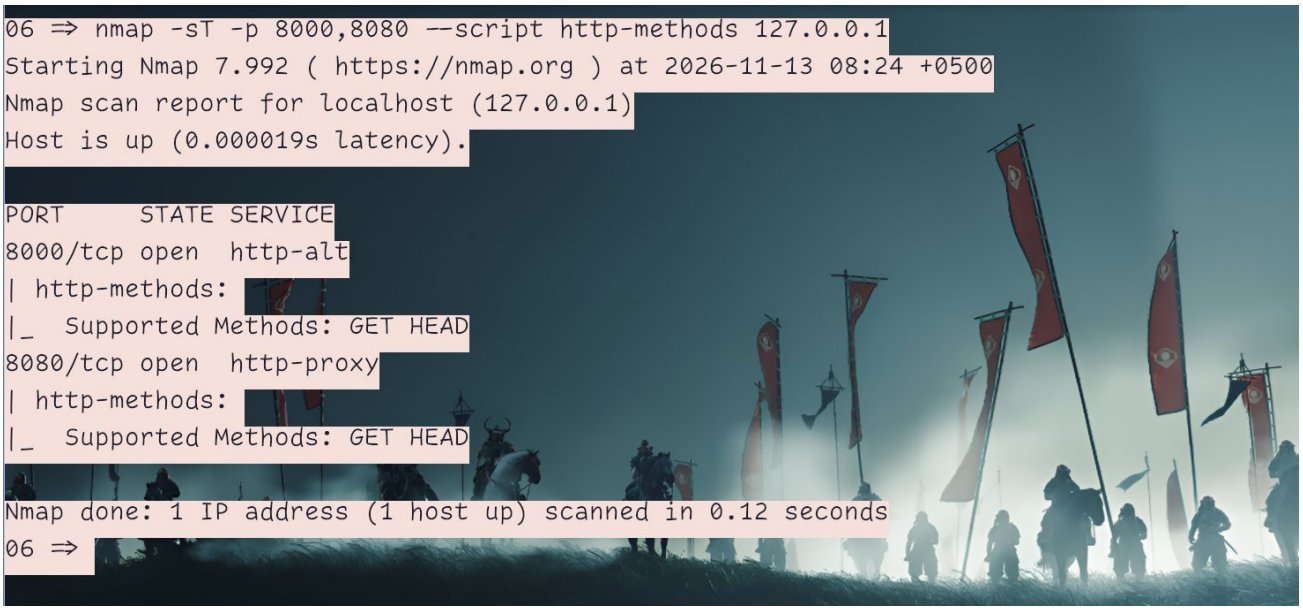
HTTP methods define the actions a client can request from a web server. Common HTTP methods include:

- GET: Requests information from a server.
- HEAD: Requests response headers.
- DELETE: Requests deletion of a resource.

Enter the following command in Zenmap:

```
nmap -sT -p 8000,8080 --script http-methods 127.0.0.1
```

Click Scan.



The http-methods script identifies HTTP methods reported or supported by a web server.

Step 9: HTTP Server Information Scanning

Enter the following command:

```
nmap -sT -p 8000,8080 --script http-headers,http-title 127.0.0.1
```

Click Scan.



Script Explanation:

- http-headers: Retrieves HTTP response headers.
- http-title: Attempts to identify the title of the web page.

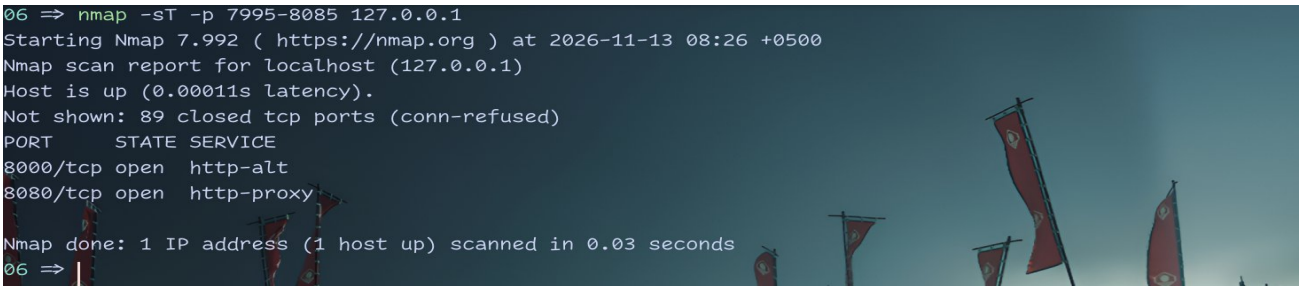
These scripts help students understand what information a web server exposes through HTTP responses..

PART D: NETWORK SCANNING

Step 10: Scan a Range of Ports

Enter the following command in Zenmap:

```
nmap -sT -p 7995-8085 127.0.0.1
```



Explanation:

This command scans TCP ports from 7995 to 8085.

It helps identify open ports within a specified range instead of scanning only individual ports.

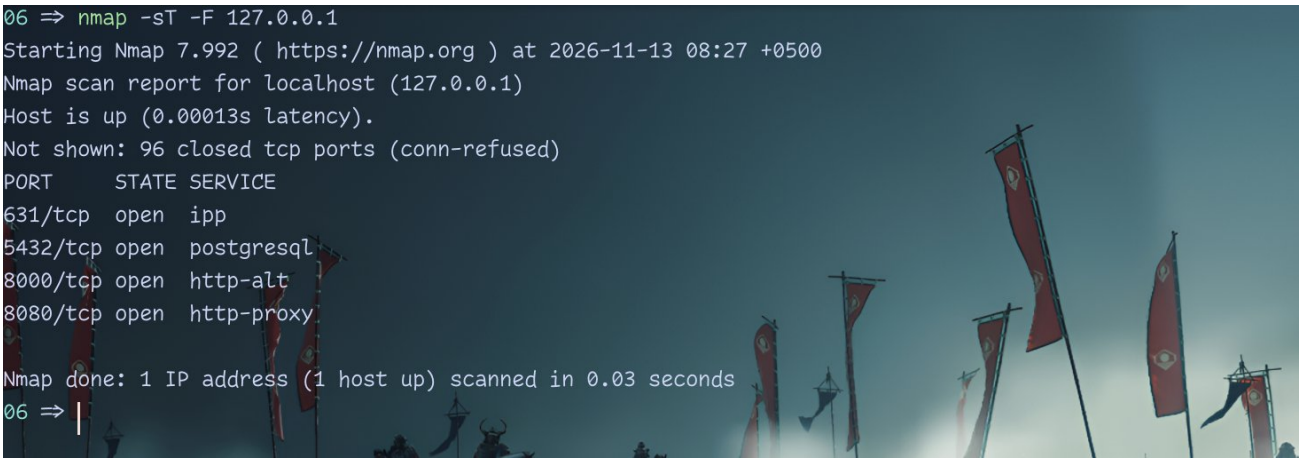
Expected Observation:

Ports 8000 and 8080 should appear as open if both web servers are running. Note that port 8080 is outside the specified range 7995–8085? No — it is within the range, so both configured ports are included.

Step 11: Perform a Fast Scan

Enter the following command:

```
nmap -sT -F 127.0.0.1
```



Explanation:

The -F option performs a fast scan by checking a smaller set of commonly used ports.

This scan is generally faster than a default port scan.

Note:

Ports 8000 and 8080 may not appear because a fast scan does not check every port.

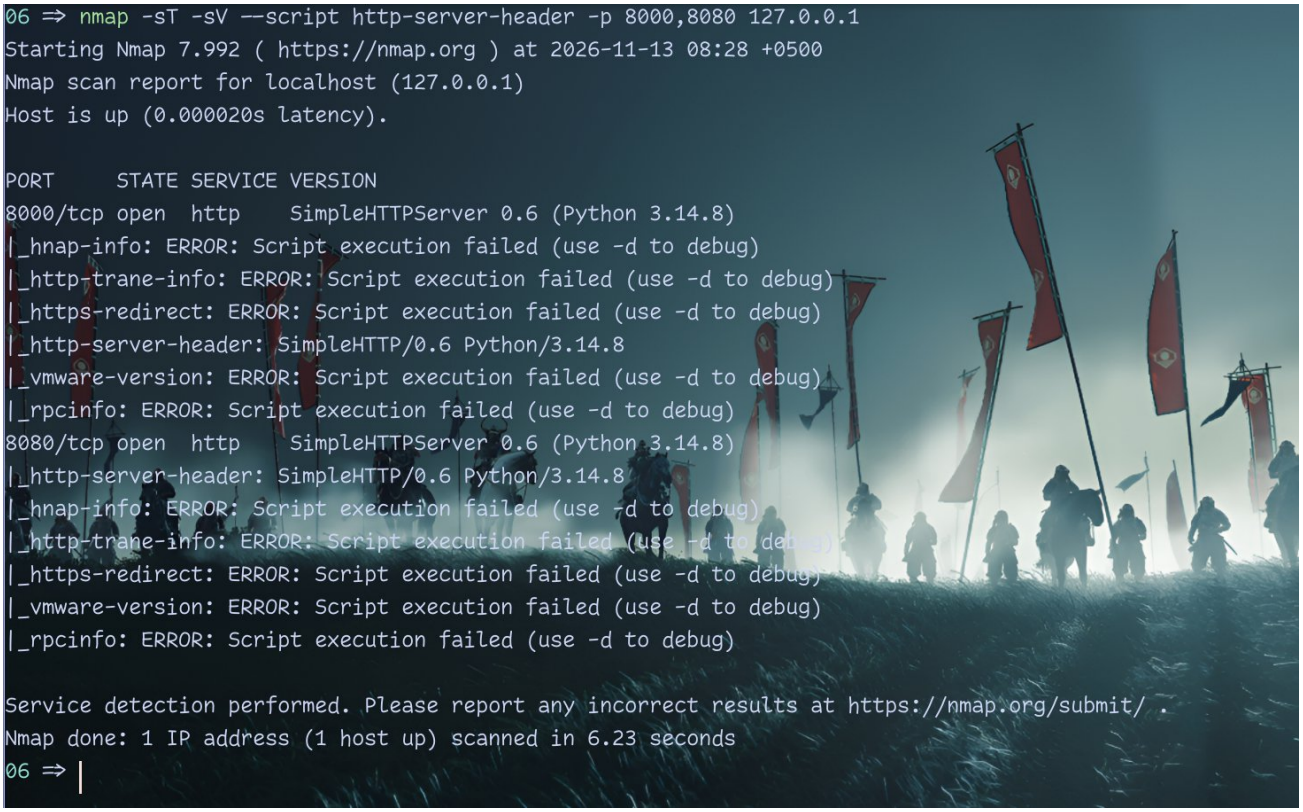
Step 12: Detect HTTP Server Information

Enter the following command:

```
nmap -sT -sV --script http-server-header -p 8000,8080 127.0.0.1
```

This command attempts to retrieve HTTP Server header information from both web servers.

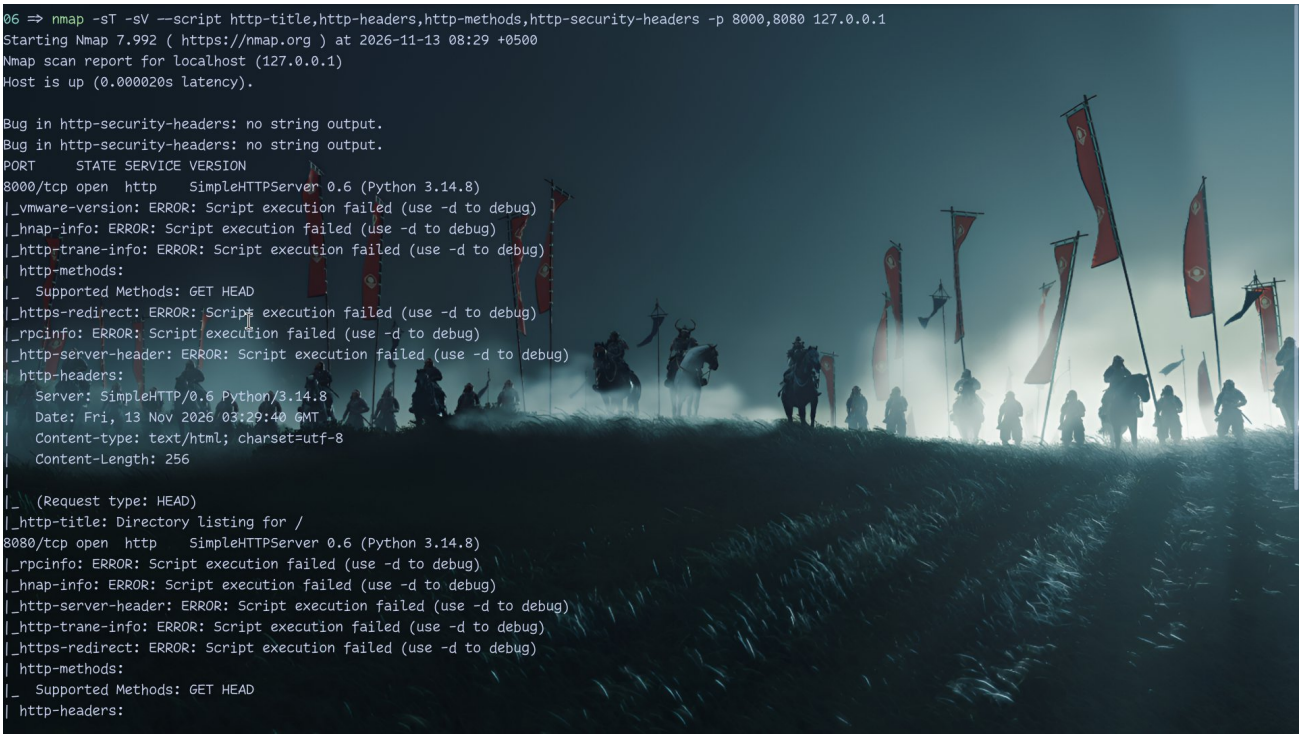
The information may reveal the web server software and version details.



Step 13: Perform Combined HTTP Security Scanning

Enter the following command:

```
nmap -sT -sV --script http-title,http-headers,http-methods,http-security-headers -p 8000,8080 127.0.0.1
```





This command performs multiple HTTP checks in a single scan.

It examines:

1. HTTP page titles.
2. HTTP response headers.
3. Supported HTTP methods.
4. HTTP security headers.
5. Available service version information.